Beam Design and AI Model Comparison

Introduction:

The goal of this project was to find a lightweight steel beam that meets the stress and deflection limits. I compared a SciPy optimizer with two AI models, GPT-4.1 mini and Llama 3.2 1B. All three methods used the provided beam analysis function to evaluate the designs.

Methods:

The beam had a span of 1 m and a center load of 10,000 N. The maximum allowed deflection was 1.0 mm, and the maximum stress was 120 MPa. The beam width and height were also limited to the ranges given in the assignment.

I used SciPy to search for a design with the lowest mass. I also tested GPT-4.1 mini and Llama 3.2 1B using an iterative process where each model suggested beam dimensions and received the results. I ran each AI model twice to compare its results.

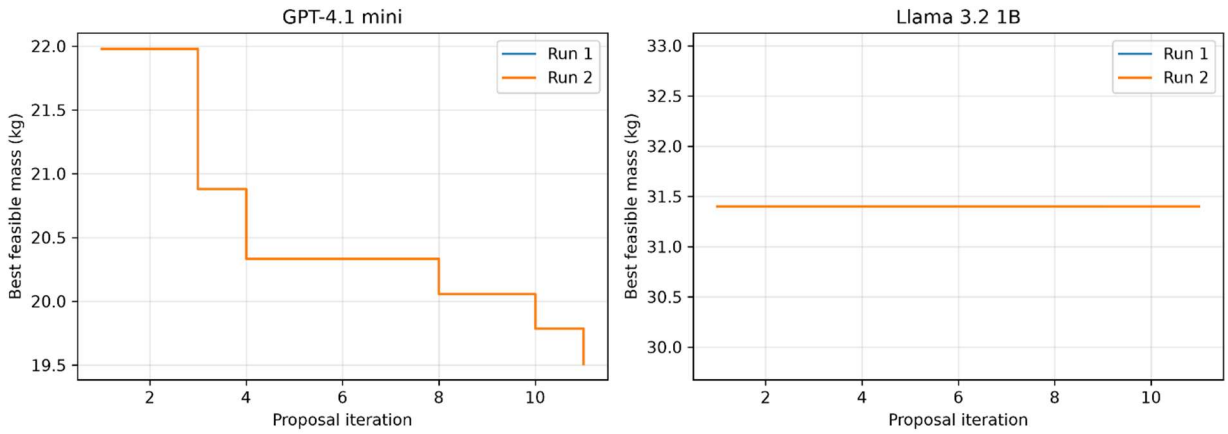
Results:

The best designs found by each method are shown below.

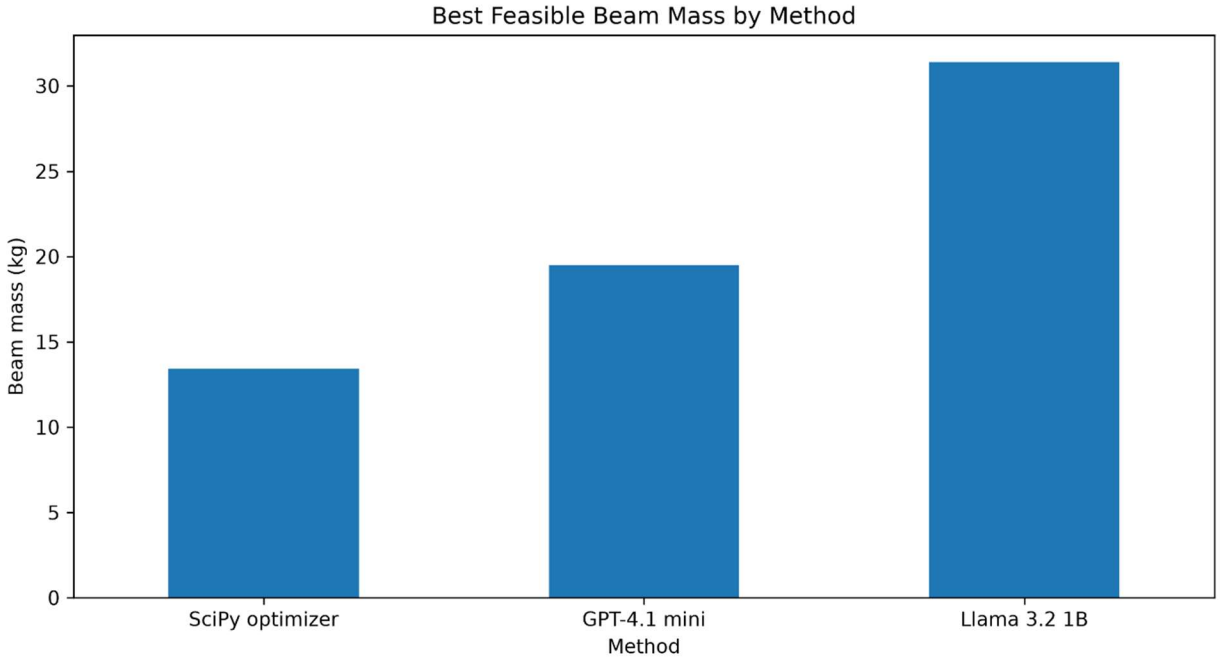
Method	**Width (mm)**	**Height (mm)**	**Mass (kg)**
SciPy optimizer	20.0	85.50	13.42
GPT-4.1 mini	35.0	71.0	19.51
Llama 3.2 1B	50.0	80.0	31.40

All three best designs met the stress and deflection limits. The SciPy optimizer found the lightest design, while GPT found a lighter design than Llama.

Search History Plot:



Beam Mass Comparison:



Discussion and Conclusion:

SciPy performed best because it found a beam with a lower mass than either AI model. GPT also improved on the starting design, but its final beam was heavier than the SciPy design. Llama repeatedly suggested the starting design and did not find a lighter solution. The SciPy and GPT designs were close to the deflection limit, while the Llama design had more room before reaching the limit.

Overall, the results show that the optimizer was more effective at finding a lightweight design in these tests. The AI results may vary depending on the suggestions made during each run.

GitHub link: https://github.com/ejeckamchizoba-cmd/beam-ai-comparison.git